keke_model

```
#load packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to becor
```

```
#load data
train <- read_csv("data-train.csv")
```

```
Rows: 89 Columns: 7
-- Column specification --------------------------------------------------------
Delimiter: ","
dbl (7): St, Re, Fr, R_moment_1, R_moment_2, R_moment_3, R_moment_4

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
#transform data
#making new variables for our summary stats

#mean = R_moment_1, C_moment_2 is variance, everything else is as commented
train <- train %>%
```

```
  mutate(
    m1 = R_moment_1, m2 = R_moment_2, m3 = R_moment_3, m4 = R_moment_4,

    # central moments
    C_moment_2 = m2 - m1^2,                                  # variance
    C_moment_3 = m3 - 3*m1*m2 + 2*m1^3,
    C_moment_4 = m4 - 4*m1*m3 + 6*m1^2*m2 - 3*m1^4,

    # standardized versions
    sd_R     = sqrt(C_moment_2),
    skewness = C_moment_3 / sd_R^3,
    kurtosis = C_moment_4 / sd_R^4,                          # raw kurtosis
    ex_kurt  = kurtosis - 3                                  # excess kurtosis
  )

train <- train %>%
  mutate(
    Re_group = factor(Re),
    Fr_group = factor(Fr)
  )

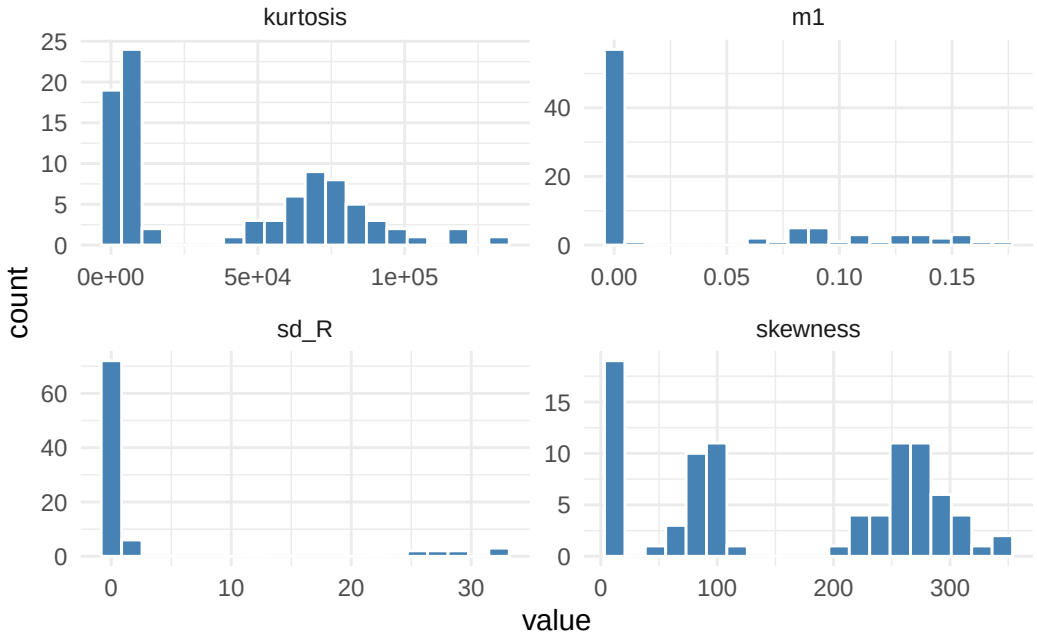
outcomes <- c("m1", "sd_R", "skewness", "kurtosis")

summary(train[outcomes])
```

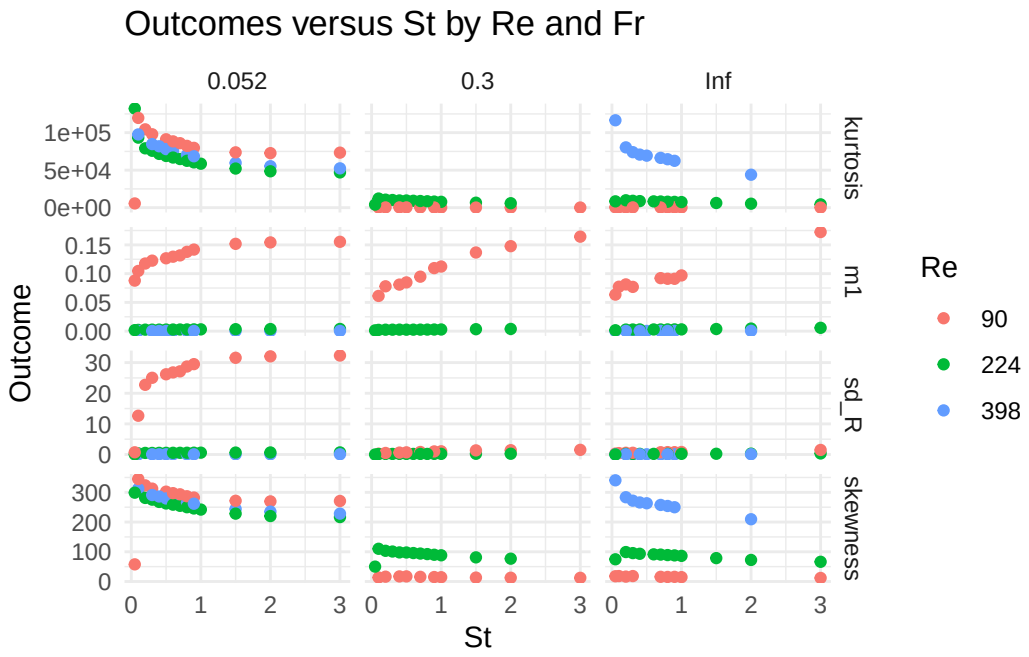
```
       m1                sd_R              skewness          kurtosis
 Min.   :0.000222   Min.   : 0.01006   Min.   : 11.97   Min.   :   150.5
 1st Qu.:0.002157   1st Qu.: 0.15643   1st Qu.: 72.55   1st Qu.:  5622.3
 Median :0.002958   Median : 0.28423   Median :110.12   Median : 12158.7
 Mean   :0.040394   Mean   : 3.65112   Mean   :162.81   Mean   : 39749.6
 3rd Qu.:0.087868   3rd Qu.: 0.72579   3rd Qu.:269.54   3rd Qu.: 72732.4
 Max.   :0.172340   Max.   :32.31526   Max.   :344.91   Max.   :132136.7
```

```
eda_data <- train %>%
  select(St, Re_group, Fr_group, all_of(outcomes)) %>%
  pivot_longer(
    cols = all_of(outcomes),
    names_to = "outcome",
    values_to = "value"
  )
```

```r
# Distributions of the outcomes
ggplot(eda_data, aes(x = value)) +
  geom_histogram(bins = 20, color = "white", fill = "steelblue") +
  facet_wrap(~ outcome, scales = "free") +
  theme_minimal()
```




```r
# Relationships with St, separated by Re and Fr
ggplot(
  eda_data,
  aes(x = St, y = value, color = Re_group)
) +
  geom_point() +
  facet_grid(outcome ~ Fr_group, scales = "free_y") +
  labs(color = "Re", y = "Outcome", title = "Outcomes versus St by Re and Fr") +
  theme_minimal()
```




```
results <- list()
fits <- list()

for (outcome in outcomes) {
  for (degree in 1:5) {

    model_formula <- as.formula(
      paste0(
        outcome,
        " ~ Re_group + Fr_group + poly(St, ",
        degree, ")"
      )
    )

    predictions <- numeric(nrow(train))

    # Leave-one-out cross-validation
    for (i in seq_len(nrow(train))) {
      fit <- lm(model_formula, data = train[-i, ])

      predictions[i] <- predict(
        fit,
        newdata = train[i, , drop = FALSE]
```

```
      )
    }

    observed <- train[[outcome]]
    model_name <- paste(outcome, degree, sep = "_degree_")

    results[[model_name]] <- data.frame(
      outcome = outcome,
      degree = degree,
      RMSE = sqrt(mean((observed - predictions)^2)),
      MAE = mean(abs(observed - predictions))
    )

    # Save the model fitted on all observations
    fits[[model_name]] <- lm(model_formula, data = train)
  }
}

cv_results <- do.call(rbind, results)
rownames(cv_results) <- NULL

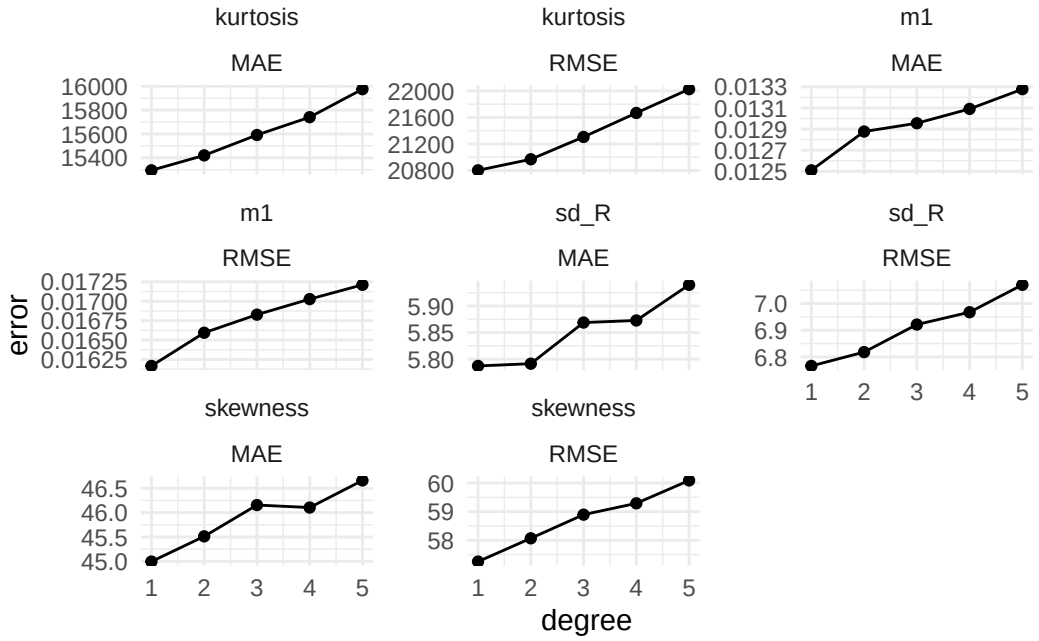
cv_results <- cv_results[
  order(cv_results$outcome, cv_results$RMSE),
]

print(cv_results)
```

```
    outcome degree         RMSE          MAE
16 kurtosis      1 2.080217e+04 1.529544e+04
17 kurtosis      2 2.096850e+04 1.542056e+04
18 kurtosis      3 2.130548e+04 1.559127e+04
19 kurtosis      4 2.166752e+04 1.574075e+04
20 kurtosis      5 2.202634e+04 1.597525e+04
1        m1      1 1.616672e-02 1.250941e-02
2        m1      2 1.659411e-02 1.287653e-02
3        m1      3 1.682715e-02 1.295516e-02
4        m1      4 1.702713e-02 1.309085e-02
5        m1      5 1.721137e-02 1.327776e-02
6      sd_R      1 6.766073e+00 5.787222e+00
7      sd_R      2 6.817985e+00 5.791457e+00
8      sd_R      3 6.921242e+00 5.868843e+00
9      sd_R      4 6.967510e+00 5.872705e+00
```

```
10      sd_R       5 7.070173e+00 5.939783e+00
11 skewness       1 5.725744e+01 4.499625e+01
12 skewness       2 5.806958e+01 4.551313e+01
13 skewness       3 5.889492e+01 4.615527e+01
14 skewness       4 5.928963e+01 4.610428e+01
15 skewness       5 6.008891e+01 4.665974e+01
```

```r
# Plot both error measures
cv_results %>%
  pivot_longer(
    cols = c(RMSE, MAE),
    names_to = "metric",
    values_to = "error"
  ) %>%
  ggplot(aes(x = degree, y = error)) +
  geom_line() +
  geom_point() +
  facet_wrap(vars(outcome, metric), scales = "free_y") +
  scale_x_continuous(breaks = 1:5) +
  theme_minimal()
```



```r
# Lowest LOOCV RMSE for each outcome
best_models <- cv_results %>%
  group_by(outcome) %>%
  slice_min(RMSE, n = 1, with_ties = FALSE) %>%
  ungroup()

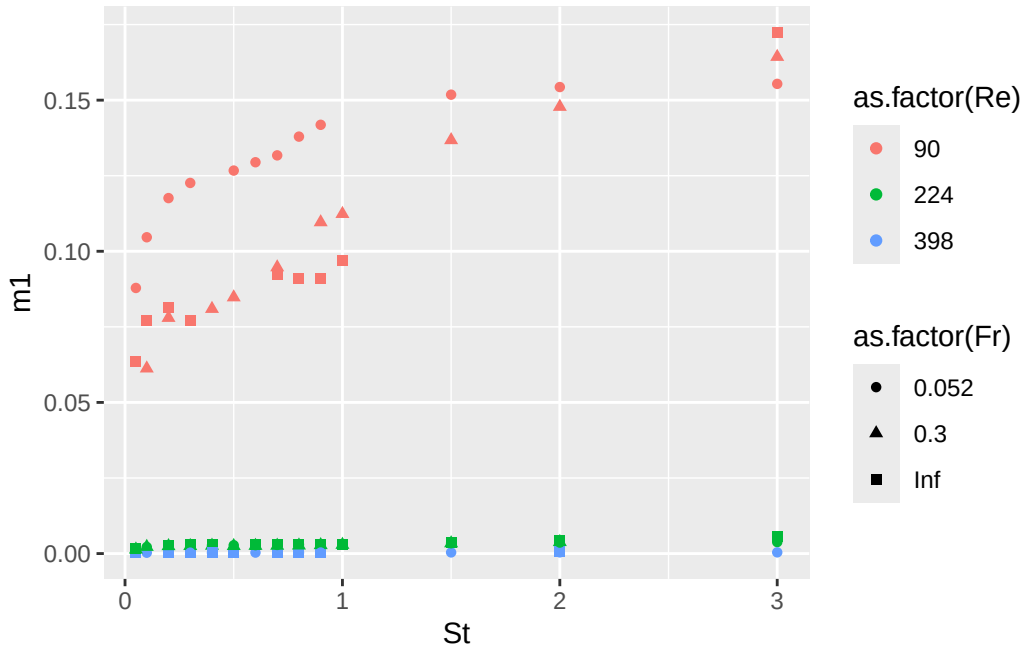
best_models
```

```
# A tibble: 4 x 4
  outcome  degree       RMSE        MAE
  <chr>     <int>      <dbl>      <dbl>
1 kurtosis      1 20802.     15295.
2 m1            1     0.0162     0.0125
3 sd_R          1     6.77       5.79
4 skewness      1    57.3       45.0
```

Linear Model Mean

```r
data <- train %>%
  mutate(
    Re = factor(Re),
    Fr = factor(Fr)
  )

train |>
  ggplot(aes(x = St, y = m1, color = as.factor(Re), shape = as.factor(Fr))) +
  geom_point()
```




```
formulas <- list(
  no_logs  = m1 ~ St + Re + Fr,
  log_mean = log(m1) ~ St + Re + Fr,
  log_St   = m1 ~ log(St) + Re + Fr,
  log_both = log(m1) ~ log(St) + Re + Fr
)

results <- list()
mean_fits <- list()

for (model_name in names(formulas)) {

  predictions <- numeric(nrow(data))
  log_response <- model_name %in% c("log_mean", "log_both")

  for (i in seq_len(nrow(data))) {

    # Fit without observation i
    fit <- lm(formulas[[model_name]], data = data[-i, ])

    pred <- predict(fit, newdata = data[i, , drop = FALSE])

    if (log_response) {
```

```
      # estimated only from this fold's training observations
      correction <- mean(exp(residuals(fit)))
      pred <- exp(pred) * correction
    }

    predictions[i] <- pred
  }

  results[[model_name]] <- tibble(
    model = model_name,
    RMSE = sqrt(mean((data$m1 - predictions)^2)),
    MAE = mean(abs(data$m1 - predictions))
  )

  mean_fits[[model_name]] <- lm(
    formulas[[model_name]],
    data = data
  )
}

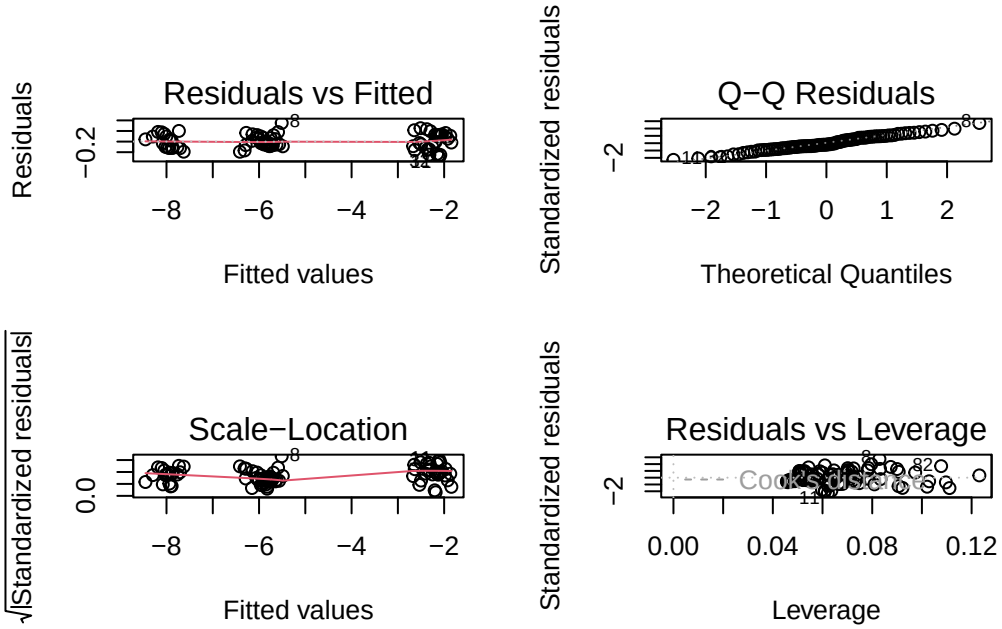
mean_comparison <- bind_rows(results) %>%
  arrange(RMSE)

mean_comparison
```

```
# A tibble: 4 x 3
  model      RMSE     MAE
  <chr>     <dbl>   <dbl>
1 log_both 0.0108 0.00572
2 log_mean 0.0116 0.00621
3 no_logs  0.0162 0.0125
4 log_St   0.0164 0.0126
```

Among the additive models, logging both mean volume and `St` produced the lowest original-scale LOOCV RMSE (0.01078) and MAE (0.00572). We therefore used both log transformations when investigating interactions.

```
par(mfrow = c(2, 2))
plot(mean_fits[["log_both"]])
```




```
par(mfrow = c(1, 1))
```

The log–log model's residuals were broadly centered around zero, with no strong overall curvature. The Q–Q plot appeared approximately linear, although some tail departures remain. Differences in residual spread across fitted values suggest that the transformation did not fully stabilize variance, motivating further investigation of interaction effects.

```
# Compare interaction effects, keeping both log transformations
formulas <- list(
  additive = log(m1) ~ log(St) + Re + Fr,
  Re_Fr = log(m1) ~ log(St) + Re * Fr,
  St_Re = log(m1) ~ log(St) * Re + Fr,
  St_Fr = log(m1) ~ log(St) * Fr + Re,
  pairwise = log(m1) ~ (log(St) + Re + Fr)^2,
  threeway = log(m1) ~ log(St) * Re * Fr
)

interaction_results <- list()
interaction_fits <- list()

for (model_name in names(formulas)) {
```

```r
  predictions <- numeric(nrow(data))

  for (i in seq_len(nrow(data))) {

    fit <- lm(formulas[[model_name]], data = data[-i, ])

    pred <- predict(fit, newdata = data[i, , drop = FALSE])

    # Back-transform using a training-fold smearing correction
    correction <- mean(exp(residuals(fit)))
    predictions[i] <- exp(pred) * correction
  }

  interaction_results[[model_name]] <- tibble(
    model = model_name,
    RMSE = sqrt(mean((data$m1 - predictions)^2)),
    MAE = mean(abs(data$m1 - predictions))
  )

  interaction_fits[[model_name]] <- lm(
    formulas[[model_name]],
    data = data
  )
}

# Retrieve the saved models
m_additive <- interaction_fits[["additive"]]
m_Re_Fr <- interaction_fits[["Re_Fr"]]
m_St_Re <- interaction_fits[["St_Re"]]
m_St_Fr <- interaction_fits[["St_Fr"]]
m_pairwise <- interaction_fits[["pairwise"]]
m_threeway <- interaction_fits[["threeway"]]

# Compare nested models using partial F-tests
anova(m_additive, m_pairwise, m_threeway)
```

```
Analysis of Variance Table

Model 1: log(m1) ~ log(St) + Re + Fr
Model 2: log(m1) ~ (log(St) + Re + Fr)^2
Model 3: log(m1) ~ log(St) * Re * Fr
  Res.Df     RSS Df Sum of Sq       F    Pr(>F)
```

```
1     83 1.47283
2     76 0.61575  7   0.85709 16.9541 4.261e-13 ***
3     73 0.52720  3   0.08855  4.0869  0.009704 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Compare all candidates: lower AIC/BIC is preferred
AIC(m_additive, m_Re_Fr, m_St_Re, m_St_Fr,
    m_pairwise, m_threeway)
```

```
           df        AIC
m_additive  7  -98.45779
m_Re_Fr    10 -147.59511
m_St_Re     9  -98.60003
m_St_Fr     9 -106.64367
m_pairwise 14 -162.07536
m_threeway 17 -169.89312
```

```r
BIC(m_additive, m_Re_Fr, m_St_Re, m_St_Fr,
    m_pairwise, m_threeway)
```

```
           df        BIC
m_additive  7  -81.03734
m_Re_Fr    10 -122.70874
m_St_Re     9  -76.20230
m_St_Fr     9  -84.24594
m_pairwise 14 -127.23445
m_threeway 17 -127.58631
```

```r
# Coefficients and model summaries
summary(m_additive)
```

```

Call:
lm(formula = formulas[[model_name]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.29320 -0.08645 -0.02696  0.10178  0.35112
```

```
Coefficients:
            Estimate Std. Error  t value Pr(>|t|)
(Intercept) -2.05888    0.03091  -66.612  < 2e-16 ***
log(St)      0.19597    0.01318   14.865  < 2e-16 ***
Re224       -3.64269    0.03206 -113.608  < 2e-16 ***
Re398       -5.78493    0.04035 -143.369  < 2e-16 ***
Fr0.3       -0.12425    0.03696   -3.362  0.00117 **
FrInf       -0.02386    0.03299   -0.723  0.47155
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1332 on 83 degrees of freedom
Multiple R-squared:  0.9967,    Adjusted R-squared:  0.9965
F-statistic:  4953 on 5 and 83 DF,  p-value: < 2.2e-16
```

```
summary(m_Re_Fr)
```

```
Call:
lm(formula = formulas[[model_name]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.20526 -0.06849 -0.01402  0.06180  0.30813

Coefficients: (1 not defined because of singularities)
             Estimate Std. Error  t value Pr(>|t|)
(Intercept) -1.922953   0.029456  -65.282  < 2e-16 ***
log(St)      0.197340   0.009959   19.815  < 2e-16 ***
Re224       -3.827987   0.039160  -97.751  < 2e-16 ***
Re398       -6.012095   0.042709 -140.770  < 2e-16 ***
Fr0.3       -0.280976   0.042707   -6.579 4.51e-09 ***
FrInf       -0.315004   0.043941   -7.169 3.36e-10 ***
Re224:Fr0.3  0.223512   0.057469    3.889 0.000207 ***
Re398:Fr0.3        NA         NA       NA       NA
Re224:FrInf  0.381988   0.058927    6.482 6.87e-09 ***
Re398:FrInf  0.485437   0.063437    7.652 3.87e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09954 on 80 degrees of freedom
Multiple R-squared:  0.9982,    Adjusted R-squared:  0.998
```

```
F-statistic:  5552 on 8 and 80 DF,  p-value: < 2.2e-16
```

```
summary(m_St_Re)
```

```
Call:
lm(formula = formulas[[model_name]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.28984 -0.08208 -0.02634  0.09815  0.32279

Coefficients:
              Estimate Std. Error  t value Pr(>|t|)
(Intercept)   -2.05417    0.03261  -62.991  < 2e-16 ***
log(St)        0.19900    0.02095    9.501 8.15e-15 ***
Re224         -3.63245    0.03650  -99.512  < 2e-16 ***
Re398         -5.81682    0.04594 -126.616  < 2e-16 ***
Fr0.3         -0.12604    0.03674   -3.431  0.00095 ***
FrInf         -0.03162    0.03288   -0.962  0.33899
log(St):Re224  0.01621    0.02906    0.558  0.57855
log(St):Re398 -0.05551    0.03709   -1.497  0.13835
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1317 on 81 degrees of freedom
Multiple R-squared:  0.9968,    Adjusted R-squared:  0.9965
F-statistic:  3618 on 7 and 81 DF,  p-value: < 2.2e-16
```

```
summary(m_St_Fr)
```

```
Call:
lm(formula = formulas[[model_name]], data = data)

Residuals:
     Min       1Q   Median       3Q      Max
-0.30620 -0.07165 -0.01382  0.09326  0.29971

Coefficients:
              Estimate Std. Error  t value Pr(>|t|)
```

```
(Intercept)   -2.09200    0.03076  -68.007  < 2e-16 ***
log(St)        0.14460    0.01953    7.406  1.1e-10 ***
Fr0.3         -0.06632    0.03990   -1.662  0.10033
FrInf          0.02537    0.03610    0.703  0.48419
Re224         -3.63995    0.03057 -119.054  < 2e-16 ***
Re398         -5.77375    0.03829 -150.806  < 2e-16 ***
log(St):Fr0.3  0.09595    0.03246    2.956  0.00408 **
log(St):FrInf  0.08100    0.02870    2.822  0.00600 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1259 on 81 degrees of freedom
Multiple R-squared:  0.9971,    Adjusted R-squared:  0.9968
F-statistic:  3961 on 7 and 81 DF,  p-value: < 2.2e-16
```

```
summary(m_pairwise)
```

```
Call:
lm(formula = formulas[[model_name]], data = data)

Residuals:
      Min        1Q    Median        3Q       Max
-0.153417 -0.069216 -0.006339  0.043031  0.247625

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error  t value Pr(>|t|)
(Intercept)   -1.95512    0.02851  -68.582  < 2e-16 ***
log(St)        0.14787    0.01803    8.200 4.56e-12 ***
Re224         -3.81564    0.03762 -101.433  < 2e-16 ***
Re398         -6.00541    0.04069 -147.594  < 2e-16 ***
Fr0.3         -0.23450    0.04043   -5.799 1.44e-07 ***
FrInf         -0.26508    0.04281   -6.191 2.80e-08 ***
log(St):Re224  0.01764    0.01993    0.885 0.378721
log(St):Re398 -0.01835    0.02640   -0.695 0.489127
log(St):Fr0.3  0.08727    0.02400    3.636 0.000501 ***
log(St):FrInf  0.07022    0.02062    3.405 0.001059 **
Re224:Fr0.3    0.23873    0.05236    4.560 1.93e-05 ***
Re398:Fr0.3         NA         NA       NA       NA
Re224:FrInf    0.37020    0.05355    6.914 1.28e-09 ***
Re398:FrInf    0.46298    0.05830    7.941 1.43e-11 ***
---
```

```
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09001 on 76 degrees of freedom
Multiple R-squared:  0.9986,    Adjusted R-squared:  0.9984
F-statistic:  4529 on 12 and 76 DF,  p-value: < 2.2e-16
```

```
summary(m_threeway)
```

```
Call:
lm(formula = formulas[[model_name]], data = data)

Residuals:
      Min        1Q    Median        3Q       Max
-0.179259 -0.054170 -0.006593  0.043172  0.282380

Coefficients: (2 not defined because of singularities)
                     Estimate Std. Error  t value Pr(>|t|)
(Intercept)         -1.962435   0.028150  -69.713  < 2e-16 ***
log(St)              0.136627   0.021230    6.435 1.14e-08 ***
Re224               -3.801015   0.038445  -98.868  < 2e-16 ***
Re398               -5.998674   0.040375 -148.575  < 2e-16 ***
Fr0.3               -0.204557   0.040257   -5.081 2.78e-06 ***
FrInf               -0.284063   0.044692   -6.356 1.59e-08 ***
log(St):Re224        0.040627   0.029804    1.363  0.17702
log(St):Re398       -0.008633   0.035614   -0.242  0.80914
log(St):Fr0.3        0.158267   0.034497    4.588 1.82e-05 ***
log(St):FrInf        0.050729   0.031637    1.603  0.11315
Re224:Fr0.3          0.168368   0.056313    2.990  0.00380 **
Re398:Fr0.3                NA         NA       NA       NA
Re224:FrInf          0.395734   0.058372    6.780 2.66e-09 ***
Re398:FrInf          0.483785   0.064712    7.476 1.35e-10 ***
log(St):Re224:Fr0.3 -0.126578   0.046583   -2.717  0.00822 **
log(St):Re398:Fr0.3        NA         NA       NA       NA
log(St):Re224:FrInf  0.036811   0.044457    0.828  0.41036
log(St):Re398:FrInf  0.022540   0.051283    0.440  0.66158
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.08498 on 73 degrees of freedom
Multiple R-squared:  0.9988,    Adjusted R-squared:  0.9986
F-statistic:  4065 on 15 and 73 DF,  p-value: < 2.2e-16
```

```r
interaction_comparison <- bind_rows(interaction_results) %>%
  arrange(RMSE)

interaction_comparison
```

```
# A tibble: 6 x 3
  model       RMSE     MAE
  <chr>      <dbl>   <dbl>
1 pairwise 0.00766 0.00333
2 threeway 0.00816 0.00304
3 Re_Fr    0.00886 0.00410
4 St_Fr    0.0104  0.00538
5 additive 0.0108  0.00572
6 St_Re    0.0108  0.00573
```

```r
final_mean <- m_pairwise

summary(final_mean)
```

```
Call:
lm(formula = formulas[[model_name]], data = data)

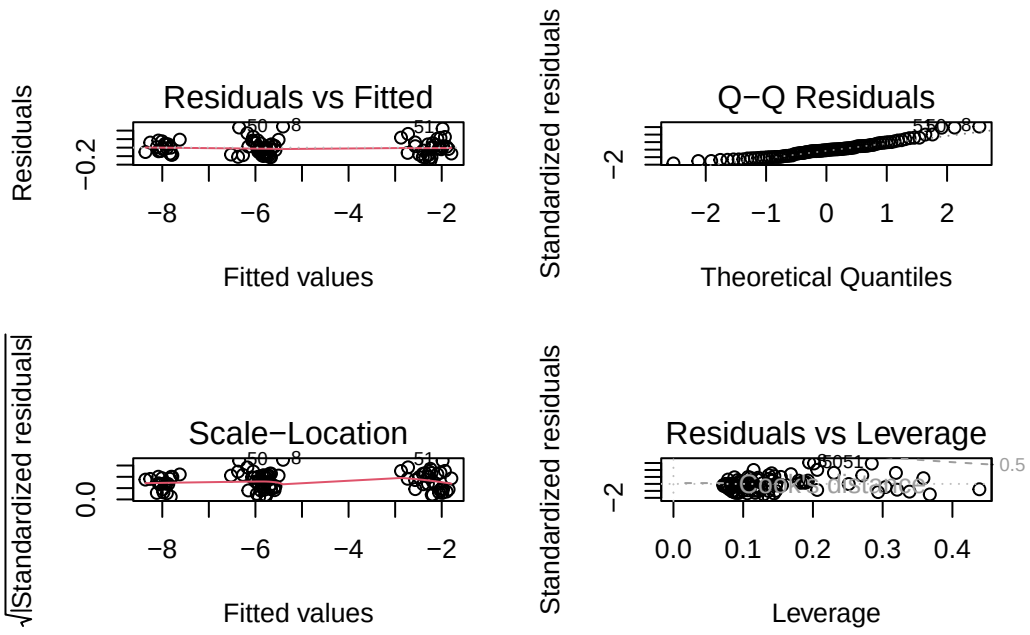
Residuals:
      Min        1Q    Median        3Q       Max
-0.153417 -0.069216 -0.006339  0.043031  0.247625

Coefficients: (1 not defined because of singularities)
              Estimate Std. Error  t value Pr(>|t|)
(Intercept)   -1.95512    0.02851  -68.582  < 2e-16 ***
log(St)        0.14787    0.01803    8.200 4.56e-12 ***
Re224         -3.81564    0.03762 -101.433  < 2e-16 ***
Re398         -6.00541    0.04069 -147.594  < 2e-16 ***
Fr0.3         -0.23450    0.04043   -5.799 1.44e-07 ***
FrInf         -0.26508    0.04281   -6.191 2.80e-08 ***
log(St):Re224  0.01764    0.01993    0.885 0.378721
log(St):Re398 -0.01835    0.02640   -0.695 0.489127
log(St):Fr0.3  0.08727    0.02400    3.636 0.000501 ***
log(St):FrInf  0.07022    0.02062    3.405 0.001059 **
Re224:Fr0.3    0.23873    0.05236    4.560 1.93e-05 ***
Re398:Fr0.3         NA         NA       NA       NA
```

```
Re224:FrInf     0.37020     0.05355     6.914 1.28e-09 ***
Re398:FrInf     0.46298     0.05830     7.941 1.43e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09001 on 76 degrees of freedom
Multiple R-squared:  0.9986,    Adjusted R-squared:  0.9984
F-statistic:  4529 on 12 and 76 DF,  p-value: < 2.2e-16
```

```r
par(mfrow = c(2, 2))
plot(final_mean)
```




```r
par(mfrow = c(1, 1))
```

The pairwise interaction model achieved the lowest original-scale LOOCV RMSE (0.00766), while the three-way model achieved the lowest MAE (0.00304). AIC favored the three-way model, whereas BIC differed by only 0.35 between the two models, offering little distinction. The nested-model ANOVA also supported adding three-way terms (p=0.0097), although this measures improvement in training fit rather than predictive performance. We selected the pairwise model because it had lower cross-validated RMSE and fewer parameters, while acknowledging that the three-way model performed better under MAE.

```r
fit <- final_mean

influence_table <- tibble(
  row = rownames(model.frame(fit)),
  cooks_distance = cooks.distance(fit),
  leverage = hatvalues(fit),
  studentized_residual = rstudent(fit)
) %>%
  arrange(desc(cooks_distance))

head(influence_table, 10)
```

```
# A tibble: 10 x 4
   row   cooks_distance leverage studentized_residual
   <chr>          <dbl>    <dbl>                <dbl>
 1 51            0.261     0.284                3.08
 2 8             0.175     0.195                3.25
 3 50            0.168     0.202                3.10
 4 86            0.102     0.368               -1.52
 5 4             0.0898    0.319                1.59
 6 17            0.0809    0.206                2.05
 7 85            0.0580    0.230                1.60
 8 87            0.0482    0.338               -1.11
 9 30            0.0440    0.293               -1.18
10 53            0.0424    0.271                1.22
```

```r
# Screening threshold, not an automatic deletion rule
4 / nobs(fit)
```

```
[1] 0.04494382
```

```r
# Inspect the three observations using their row names
data[match(c("51", "8", "50"), rownames(data)),
     c("St", "Re", "Fr", "m1")]
```

```
# A tibble: 3 x 4
     St Re    Fr          m1
  <dbl> <fct> <fct>    <dbl>
1   3   90    Inf    0.172
2   3   224   Inf    0.00575
3   0.1 224   0.3    0.00222
```

```r
mean_formula <- formula(final_mean)
fit_no51 <- lm(mean_formula, data = data[-51, ])
fit_no8  <- lm(mean_formula, data = data[-8, ])
fit_no50 <- lm(mean_formula, data = data[-50, ])

# Match coefficients by name
terms <- names(coef(final_mean))

coef_comparison <- cbind(
  full = coef(final_mean)[terms],
  without_51 = coef(fit_no51)[terms],
  without_8 = coef(fit_no8)[terms],
  without_50 = coef(fit_no50)[terms]
)

round(coef_comparison, 4)
```

```
                full without_51 without_8 without_50
(Intercept)   -1.9551    -1.9616   -1.9525    -1.9583
log(St)        0.1479     0.1378    0.1519     0.1431
Re224         -3.8156    -3.8060   -3.8231    -3.8096
Re398         -6.0054    -5.9956   -6.0054    -6.0023
Fr0.3         -0.2345    -0.2313   -0.2346    -0.2259
FrInf         -0.2651    -0.3166   -0.2762    -0.2661
log(St):Re224  0.0176     0.0327    0.0059     0.0271
log(St):Re398 -0.0184     0.0005   -0.0154    -0.0135
log(St):Fr0.3  0.0873     0.0886    0.0900     0.1064
log(St):FrInf  0.0702     0.0527    0.0562     0.0702
Re224:Fr0.3    0.2387     0.2372    0.2398     0.2225
Re398:Fr0.3        NA         NA        NA         NA
Re224:FrInf    0.3702     0.4127    0.3502     0.3705
Re398:FrInf    0.4630     0.5041    0.4657     0.4640
```

```r
# Calculate changes relative to full-model standard errors
full_se <- sqrt(diag(vcov(final_mean)))[terms]

changes <- sweep(
  coef_comparison[, -1, drop = FALSE],
  1, coef(final_mean)[terms], "-"
)
```

```r
changes_in_se <- sweep(changes, 1, full_se, "/")

round(changes_in_se, 2)
```

```
              without_51 without_8 without_50
(Intercept)        -0.23      0.09      -0.11
log(St)            -0.56      0.22      -0.27
Re224               0.26     -0.20       0.16
Re398               0.24      0.00       0.08
Fr0.3               0.08      0.00       0.21
FrInf              -1.20     -0.26      -0.02
log(St):Re224       0.76     -0.59       0.48
log(St):Re398       0.71      0.11       0.18
log(St):Fr0.3       0.06      0.11       0.80
log(St):FrInf      -0.85     -0.68       0.00
Re224:Fr0.3        -0.03      0.02      -0.31
Re398:Fr0.3           NA        NA         NA
Re224:FrInf         0.79     -0.37       0.01
Re398:FrInf         0.70      0.05       0.02
```

Observations 51, 8, and 50 were investigated because of their relatively large Cook's distances and studentized residuals. Removing each observation separately produced mostly modest coefficient changes. Observation 51 had the greatest effect, shifting the `FrInf` coefficient by approximately 1.2 full-model standard errors; all other changes were less than one standard error. These checks suggested some sensitivity in individual effects but did not indicate that one observation dominated the coefficient estimates. All observations were retained because no data error or substantive reason for exclusion was identified.

```r
table(data$Re, data$Fr)
```

```
     0.052 0.3 Inf
  90     12  10   9
  224    14  13  12
  398    10   0   9
```

```r
# Any NA coefficients are not independently estimable
coef(final_mean)
```

```
  (Intercept)       log(St)          Re224          Re398          Fr0.3
  -1.95512133    0.14787393   -3.81563899   -6.00541333   -0.23449558
        FrInf log(St):Re224 log(St):Re398 log(St):Fr0.3 log(St):FrInf
  -0.26508184    0.01764458   -0.01835144    0.08726689    0.07022261
  Re224:Fr0.3   Re398:Fr0.3   Re224:FrInf   Re398:FrInf
   0.23872815            NA    0.37020408    0.46298101
```

```
# Shows dependencies among model coefficients
alias(final_mean)
```

```
Model :
log(m1) ~ (log(St) + Re + Fr)^2

Complete :
            (Intercept) log(St) Re224 Re398 Fr0.3 FrInf log(St):Re224
Re398:Fr0.3 0           0       0     0     0     0     0
            log(St):Re398 log(St):Fr0.3 log(St):FrInf Re224:Fr0.3 Re224:FrInf
Re398:Fr0.3 0             0             0           0            0
            Re398:FrInf
Re398:Fr0.3 0
```

The interaction for `Re = 398` and `Fr = 0.3` was not estimatable because no training simulations were available at that combination.

Numerical Mean Model

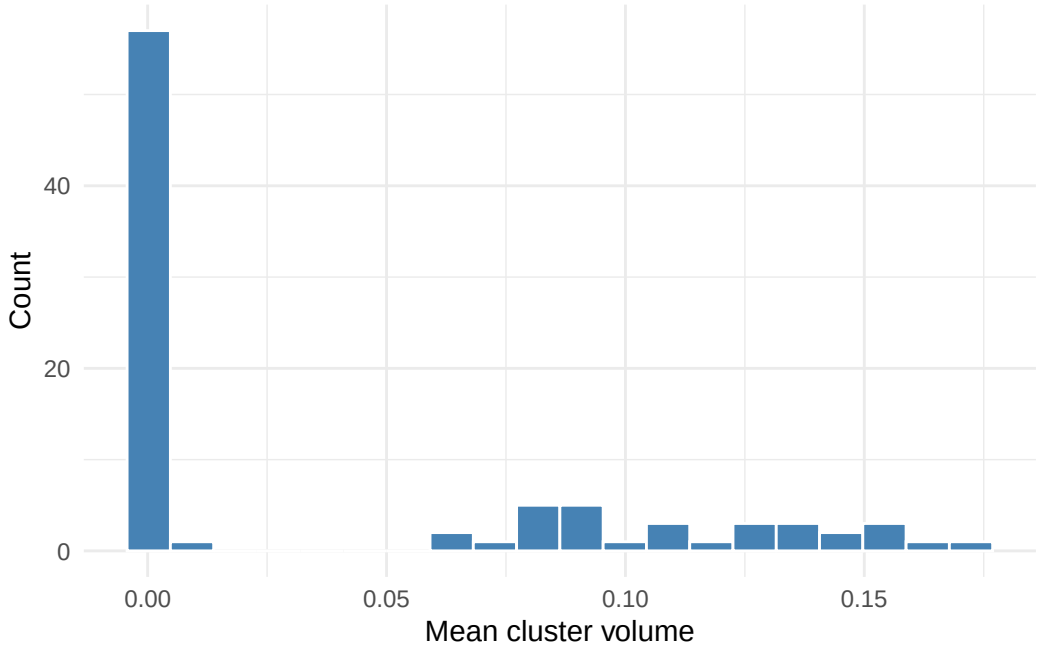
```
data_numeric <- train %>%
  mutate(
    Re = as.numeric(as.character(Re)),
    Fr = as.numeric(as.character(Fr))
  )

summary(data_numeric)
```

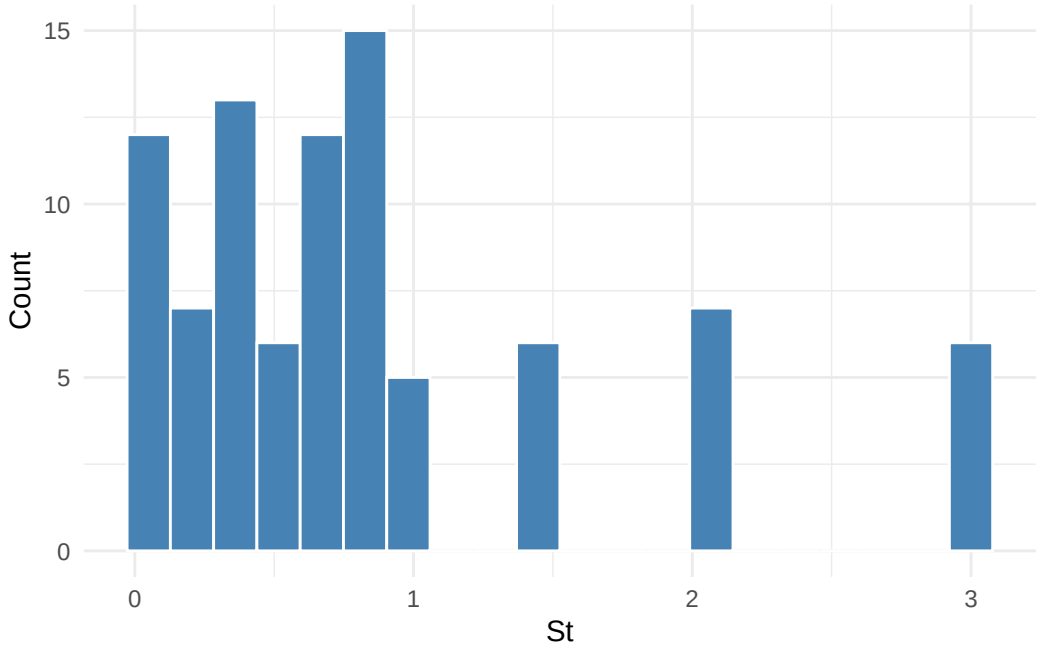
```
       St               Re              Fr          R_moment_1
 Min.   :0.0500   Min.   : 90.0   Min.   :0.052   Min.   :0.000222
 1st Qu.:0.3000   1st Qu.: 90.0   1st Qu.:0.052   1st Qu.:0.002157
 Median :0.7000   Median :224.0   Median :0.300   Median :0.002958
 Mean   :0.8596   Mean   :214.5   Mean   :  Inf   Mean   :0.040394
```

```
 3rd Qu.:1.0000   3rd Qu.:224.0   3rd Qu.:  Inf   3rd Qu.:0.087868
 Max.   :3.0000   Max.   :398.0   Max.   :  Inf   Max.   :0.172340
   R_moment_2          R_moment_3          R_moment_4              m1
 Min.   :1.000e-04   Min.   :      0.0   Min.   :0.000e+00   Min.   :0.000222
 1st Qu.:2.448e-02   1st Qu.:      0.2   1st Qu.:3.000e+00   1st Qu.:0.002157
 Median :8.080e-02   Median :      1.2   Median :2.100e+01   Median :0.002958
 Mean   :9.249e+01   Mean   : 753369.7   Mean   :6.194e+09   Mean   :0.040394
 3rd Qu.:5.345e-01   3rd Qu.:     40.5   3rd Qu.:5.345e+03   3rd Qu.:0.087868
 Max.   :1.044e+03   Max.   :9140000.0   Max.   :8.000e+10   Max.   :0.172340
       m2                  m3                  m4
 Min.   :1.000e-04   Min.   :      0.0   Min.   :0.000e+00
 1st Qu.:2.448e-02   1st Qu.:      0.2   1st Qu.:3.000e+00
 Median :8.080e-02   Median :      1.2   Median :2.100e+01
 Mean   :9.249e+01   Mean   : 753369.7   Mean   :6.194e+09
 3rd Qu.:5.345e-01   3rd Qu.:     40.5   3rd Qu.:5.345e+03
 Max.   :1.044e+03   Max.   :9140000.0   Max.   :8.000e+10
   C_moment_2          C_moment_3          C_moment_4              sd_R
 Min.   :1.000e-04   Min.   :      0.0   Min.   :0.000e+00   Min.   : 0.01006
 1st Qu.:2.447e-02   1st Qu.:      0.2   1st Qu.:3.000e+00   1st Qu.: 0.15643
 Median :8.078e-02   Median :      1.2   Median :2.000e+01   Median : 0.28423
 Mean   :9.249e+01   Mean   : 753331.2   Mean   :6.194e+09   Mean   : 3.65112
 3rd Qu.:5.268e-01   3rd Qu.:     39.3   3rd Qu.:5.345e+03   3rd Qu.: 0.72579
 Max.   :1.044e+03   Max.   :9139513.2   Max.   :7.999e+10   Max.   :32.31526
    skewness          kurtosis            ex_kurt          Re_group  Fr_group
 Min.   : 11.97   Min.   :   150.5   Min.   :   147.5   90 :31   0.052:36
 1st Qu.: 72.55   1st Qu.:  5622.3   1st Qu.:  5619.3   224:39   0.3  :23
 Median :110.12   Median : 12158.7   Median : 12155.7   398:19   Inf  :30
 Mean   :162.81   Mean   : 39749.6   Mean   : 39746.6
 3rd Qu.:269.54   3rd Qu.: 72732.4   3rd Qu.: 72729.4
 Max.   :344.91   Max.   :132136.7   Max.   :132133.7
```

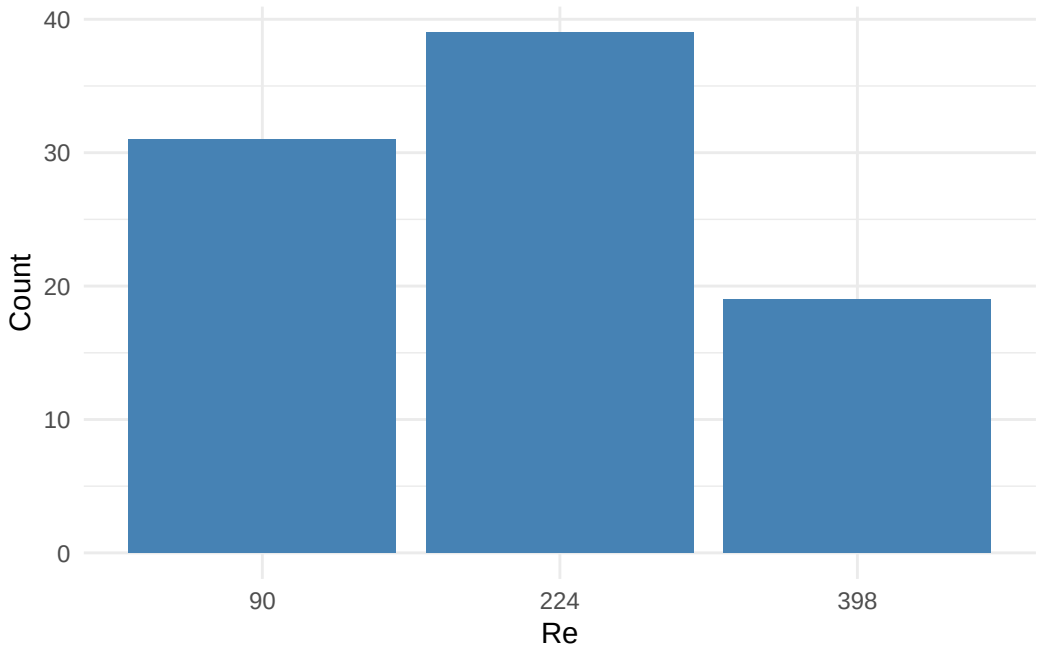
```r
# Mean volume
ggplot(data_numeric, aes(x = m1)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  labs(x = "Mean cluster volume", y = "Count") +
  theme_minimal()
```



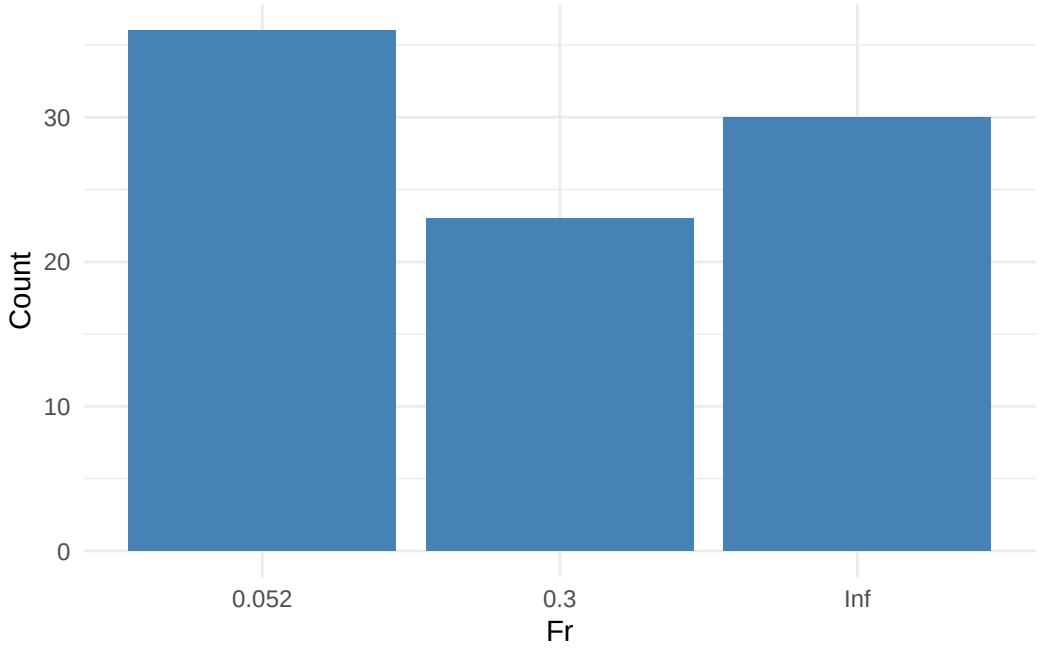

```
# St
ggplot(data_numeric, aes(x = St)) +
  geom_histogram(bins = 20, fill = "steelblue", color = "white") +
  labs(x = "St", y = "Count") +
  theme_minimal()
```




```
# Re
ggplot(data_numeric, aes(x = factor(Re))) +
  geom_bar(fill = "steelblue") +
  labs(x = "Re", y = "Count") +
  theme_minimal()
```

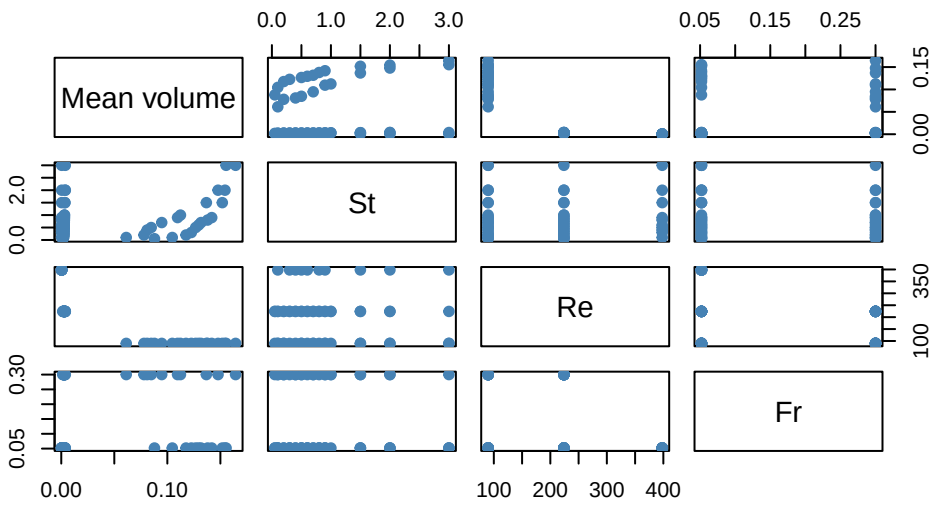



```
# Fr, including infinity
ggplot(data_numeric, aes(x = factor(Fr))) +
  geom_bar(fill = "steelblue") +
  labs(x = "Fr", y = "Count") +
  theme_minimal()
```

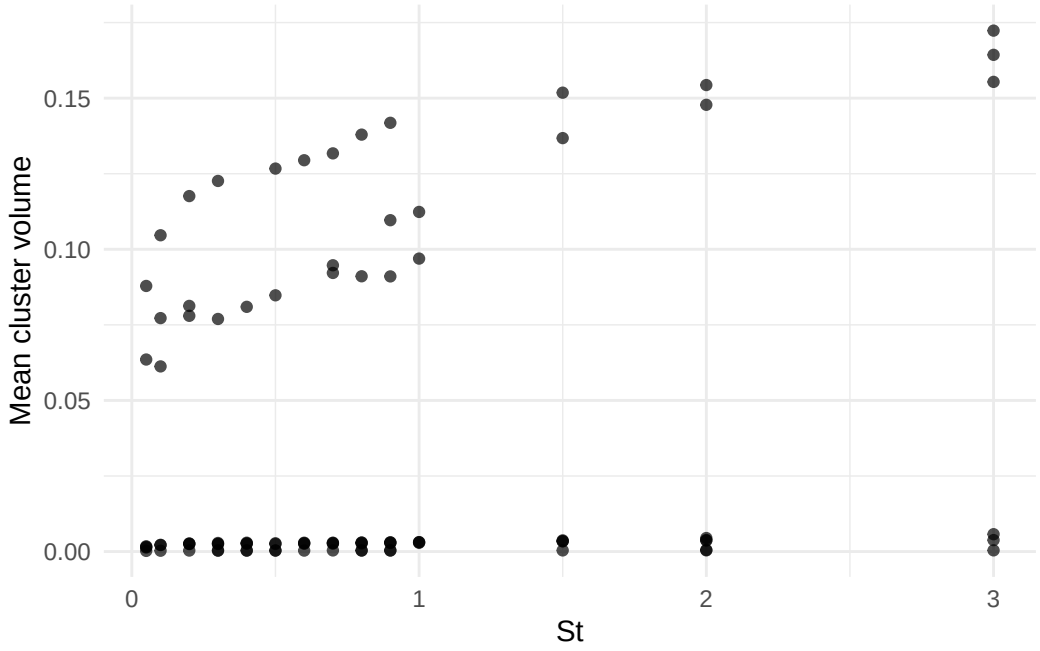



```
finite_data <- data_numeric %>%
  filter(is.finite(Fr)) %>%
  select(m1, St, Re, Fr)

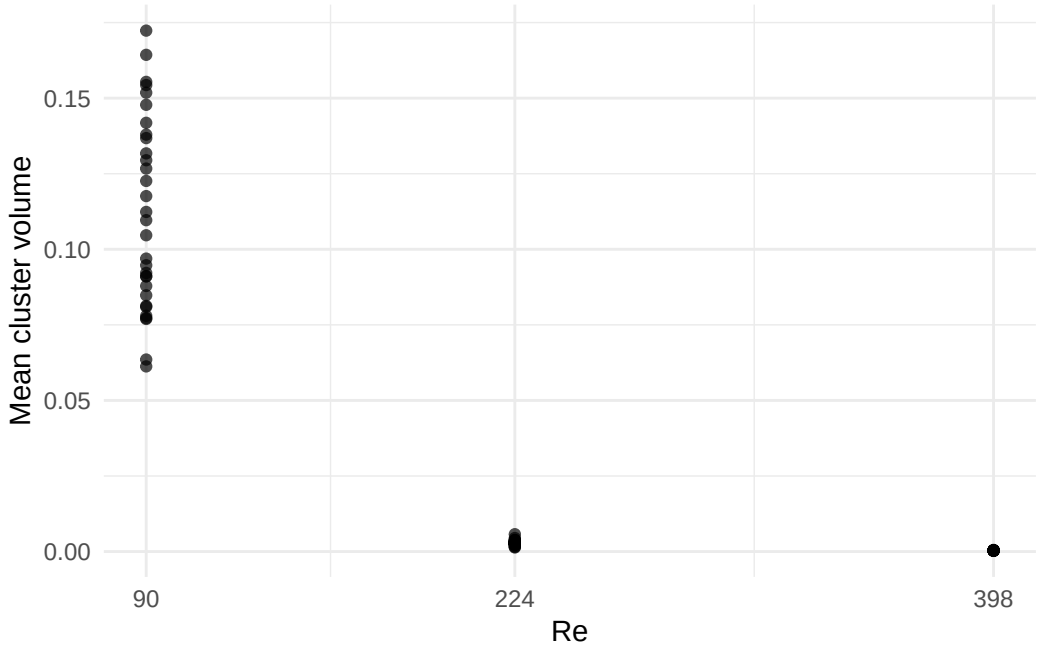
pairs(
  finite_data,
  labels = c("Mean volume", "St", "Re", "Fr"),
  main = "Pairwise relationships: finite Fr observations only",
  pch = 19,
  col = "steelblue"
)
```



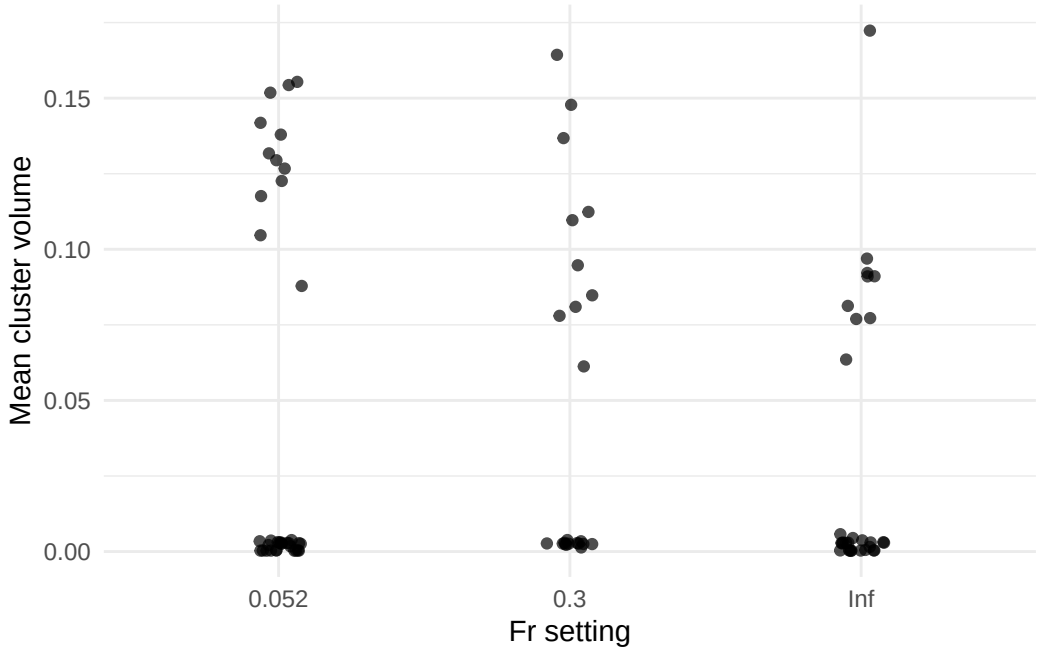

```r
# Mean versus St
ggplot(data_numeric, aes(x = St, y = m1)) +
  geom_point(alpha = 0.7) +
  labs(x = "St", y = "Mean cluster volume") +
  theme_minimal()
```



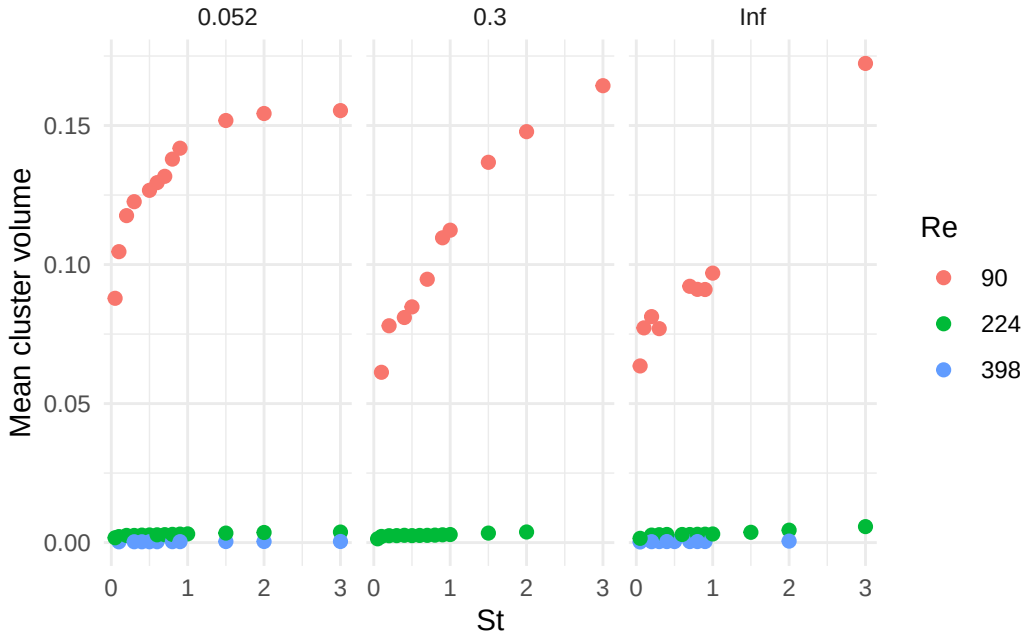

```
# Mean versus Re
ggplot(data_numeric, aes(x = Re, y = m1)) +
  geom_point(alpha = 0.7) +
  scale_x_continuous(breaks = sort(unique(data_numeric$Re))) +
  labs(x = "Re", y = "Mean cluster volume") +
  theme_minimal()
```



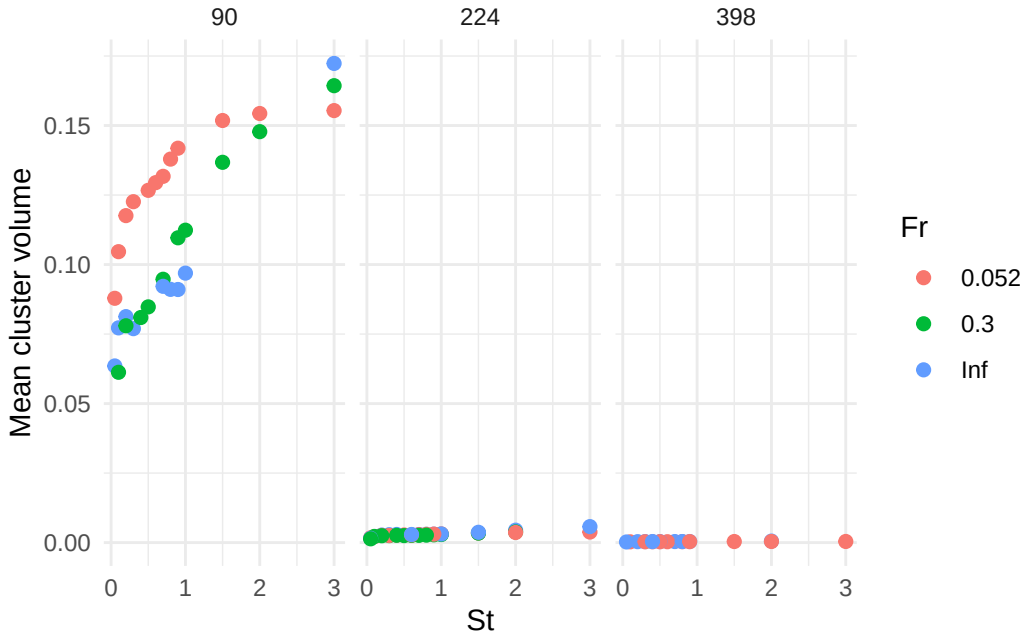

```
# Mean across Fr settings, including infinity
ggplot(data_numeric, aes(x = factor(Fr), y = m1)) +
  geom_point(
    position = position_jitter(width = 0.08, height = 0, seed = 325),
    alpha = 0.7
  ) +
  labs(x = "Fr setting", y = "Mean cluster volume") +
  theme_minimal()
```




```
ggplot(
  data_numeric,
  aes(x = St, y = m1, color = factor(Re))
) +
  geom_point(size = 2) +
  facet_wrap(~ Fr) +
  labs(
    x = "St",
    y = "Mean cluster volume",
    color = "Re"
  ) +
  theme_minimal()
```




```
ggplot(
  data_numeric,
  aes(x = St, y = m1, color = factor(Fr))
) +
  geom_point(size = 2) +
  facet_wrap(~ Re) +
  labs(
    x = "St",
    y = "Mean cluster volume",
    color = "Fr"
  ) +
  theme_minimal()
```



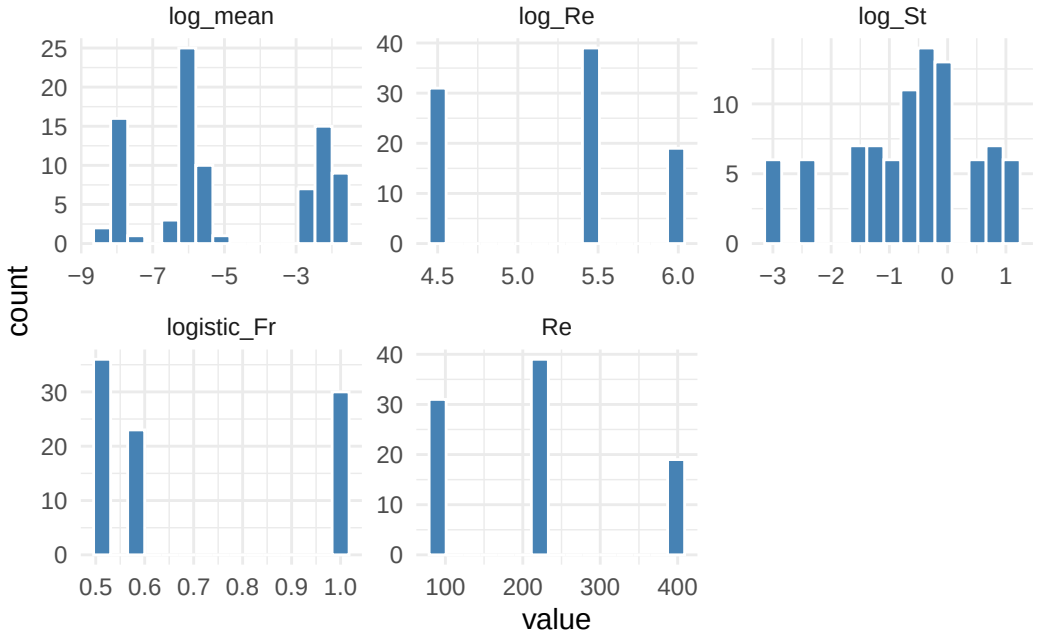

```
data_numeric <- data_numeric %>%
  mutate(
    m1 = R_moment_1,
    log_mean = log(m1),
    log_St = log(St),
    logistic_Fr = plogis(Fr),
    log_Re = log(Re)
  )

# Check the transformed variables
summary(
  data_numeric %>%
    select(log_mean, log_St, logistic_Fr, Re, log_Re)
)
```

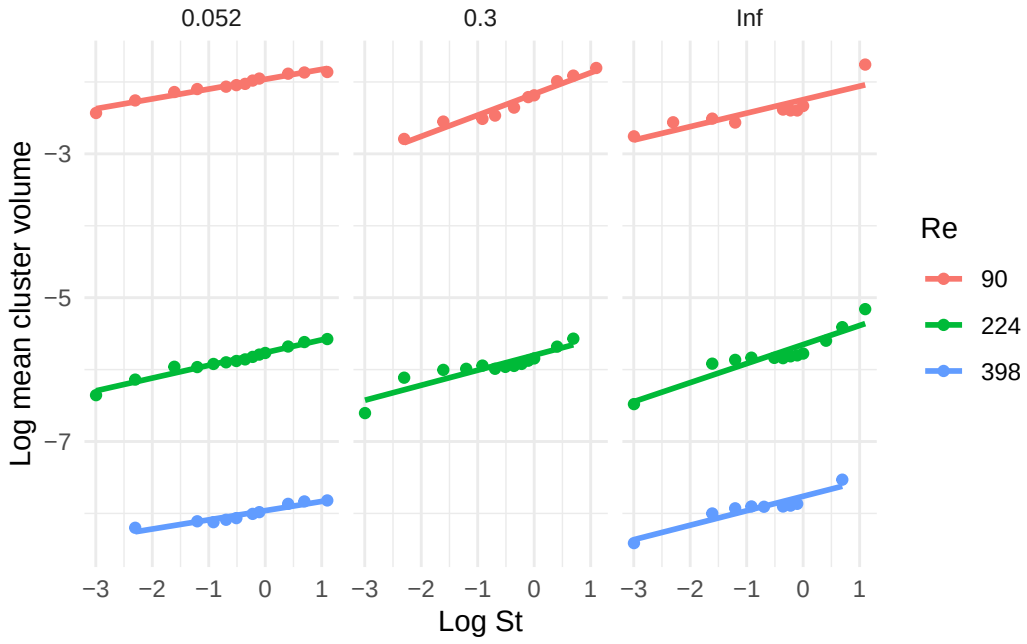
```
    log_mean          log_St         logistic_Fr          Re       
 Min.   :-8.413   Min.   :-2.9957   Min.   :0.5130   Min.   : 90.0  
 1st Qu.:-6.139   1st Qu.:-1.2040   1st Qu.:0.5130   1st Qu.: 90.0  
 Median :-5.823   Median :-0.3567   Median :0.5744   Median :224.0  
 Mean   :-5.051   Mean   :-0.6148   Mean   :0.6930   Mean   :214.5  
 3rd Qu.:-2.432   3rd Qu.: 0.0000   3rd Qu.:1.0000   3rd Qu.:224.0  
 Max.   :-1.758   Max.   : 1.0986   Max.   :1.0000   Max.   :398.0  
```

```
     log_Re
 Min.   :4.500
 1st Qu.:4.500
 Median :5.412
 Mean   :5.217
 3rd Qu.:5.412
 Max.   :5.986
```

```r
data_numeric %>%
  select(log_mean, log_St, logistic_Fr, Re, log_Re) %>%
  pivot_longer(
    everything(),
    names_to = "variable",
    values_to = "value"
  ) %>%
  ggplot(aes(x = value)) +
  geom_histogram(
    bins = 15,
    fill = "steelblue",
    color = "white"
  ) +
  facet_wrap(~ variable, scales = "free") +
  theme_minimal()
```

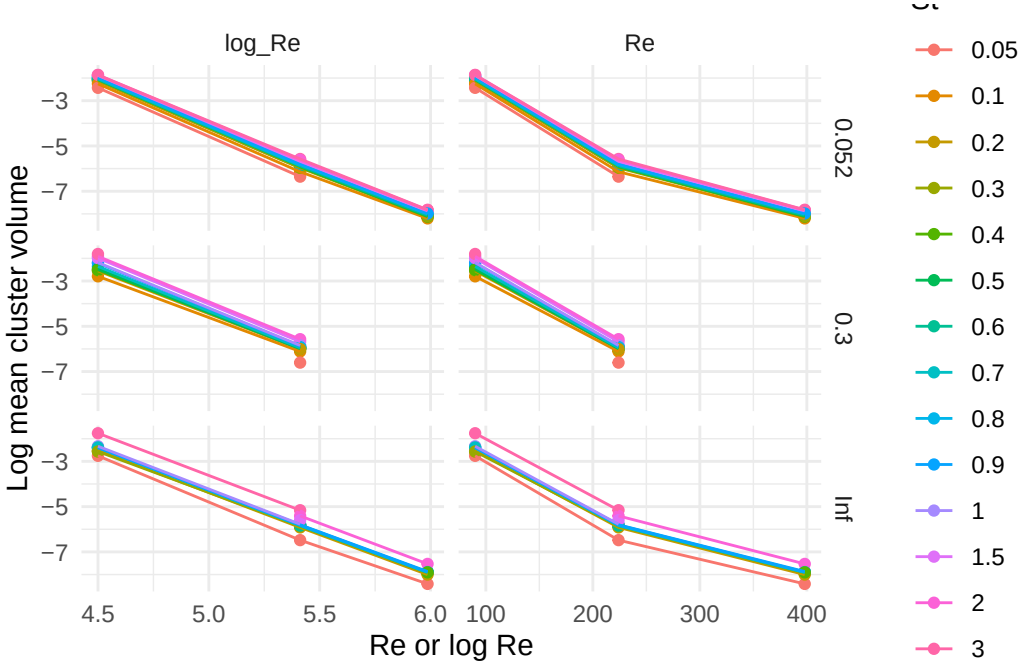


```
ggplot(
  data_numeric,
  aes(
    x = log_St,
    y = log_mean,
    color = factor(Re),
    group = factor(Re)
  )
) +
  geom_point() +
  geom_smooth(method = "lm", formula = y ~ x, se = FALSE) +
  facet_wrap(~ Fr) +
  labs(
    x = "Log St",
    y = "Log mean cluster volume",
    color = "Re"
  ) +
  theme_minimal()
```

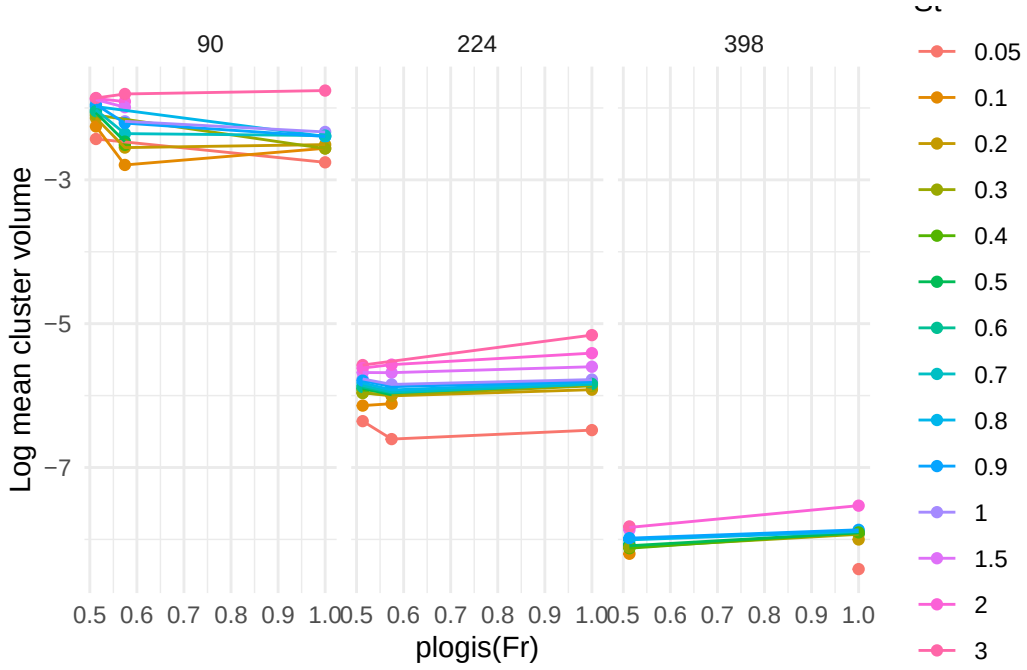



```
data_numeric %>%
  select(log_mean, St, Fr, Re, log_Re) %>%
  pivot_longer(
```

```
  cols = c(Re, log_Re),
  names_to = "Re_scale",
  values_to = "Re_value"
) %>%
ggplot(
  aes(
    x = Re_value,
    y = log_mean,
    color = factor(St),
    group = factor(St)
  )
) +
geom_point() +
geom_line() +
facet_grid(Fr ~ Re_scale, scales = "free_x") +
labs(
  x = "Re or log Re",
  y = "Log mean cluster volume",
  color = "St"
) +
theme_minimal()
```



```
ggplot(
  data_numeric,
  aes(
    x = logistic_Fr,
    y = log_mean,
    color = factor(St),
    group = factor(St)
  )
) +
  geom_point() +
  geom_line() +
  facet_wrap(~ Re) +
  labs(
    x = "plogis(Fr)",
    y = "Log mean cluster volume",
    color = "St"
  ) +
  theme_minimal()
```




```
baseline_numeric <- lm(
  log_mean ~ log_St + log_Re + logistic_Fr,
  data = data_numeric
```

```
)

summary(baseline_numeric)
```

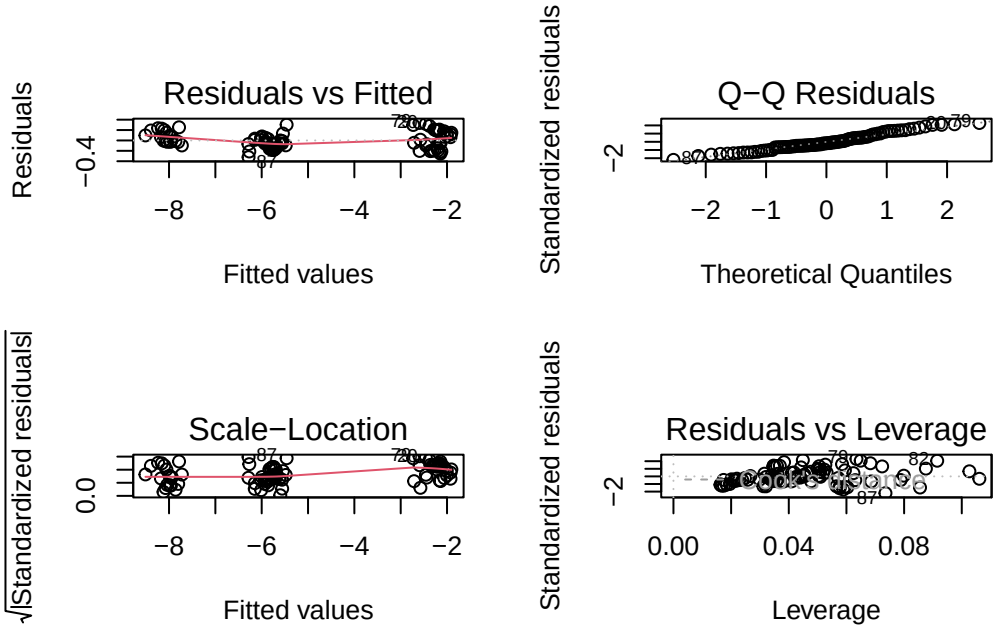
```
Call:
lm(formula = log_mean ~ log_St + log_Re + logistic_Fr, data = data_numeric)

Residuals:
     Min       1Q   Median       3Q      Max
-0.32136 -0.09344 -0.02608  0.09014  0.33613

Coefficients:
            Estimate Std. Error  t value Pr(>|t|)
(Intercept) 15.38912    0.15316  100.479   <2e-16 ***
log_St       0.19661    0.01500   13.108   <2e-16 ***
log_Re      -3.90003    0.02845 -137.106   <2e-16 ***
logistic_Fr  0.03825    0.07341    0.521    0.604
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1516 on 85 degrees of freedom
Multiple R-squared:  0.9956,    Adjusted R-squared:  0.9954
F-statistic:  6369 on 3 and 85 DF,  p-value: < 2.2e-16
```

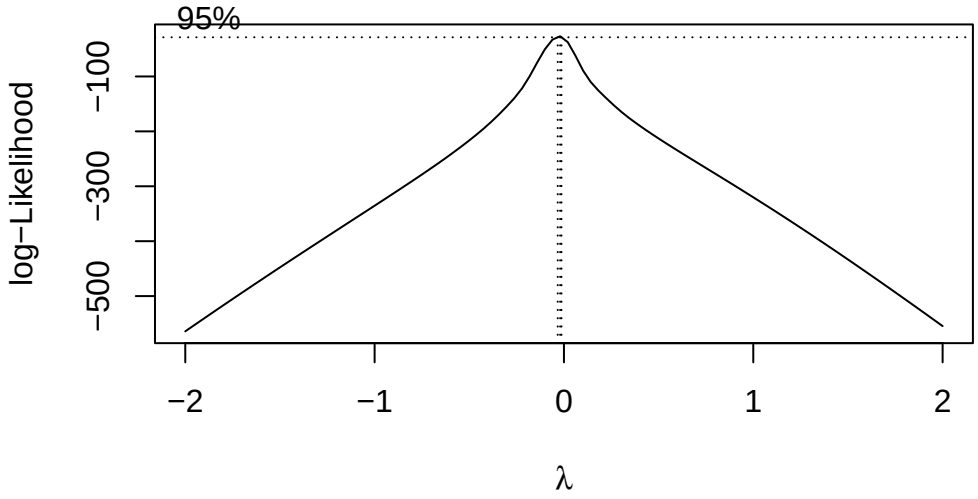
```
par(mfrow = c(2, 2))
plot(baseline_numeric)
```




```
par(mfrow = c(1, 1))
```

```
# Use the original positive outcome for Box-Cox
bc_baseline <- lm(
  m1 ~ log_St + log_Re + logistic_Fr,
  data = data_numeric
)

bc <- MASS::boxcox(
  bc_baseline,
  lambda = seq(-2, 2, by = 0.1)
)
```




```
best_lambda <- bc$x[which.max(bc$y)]
best_lambda
```

```
[1] -0.02020202
```

Although the Box–Cox confidence interval excluded zero, the estimated of approximately −0.0202 was close to zero. We therefore retained the log transformation for its simplicity and interpretable coefficients, while recognizing that Box–Cox provided a slightly better fit under the baseline model.

```
numeric_formulas <- list(
  additive = log_mean ~ log_St + log_Re + logistic_Fr,

  Re_Fr = log_mean ~ log_St + log_Re * logistic_Fr,

  St_Fr = log_mean ~ log_Re + log_St * logistic_Fr,

  St_Re = log_mean ~ logistic_Fr + log_St * log_Re,

  pairwise = log_mean ~ (log_St + log_Re + logistic_Fr)^2,

  threeway = log_mean ~ log_St * log_Re * logistic_Fr
)
```

```r
numeric_results <- list()
numeric_fits <- list()

for (model_name in names(numeric_formulas)) {

  predictions <- rep(NA_real_, nrow(data_numeric))

  for (i in seq_len(nrow(data_numeric))) {

    fit <- lm(
      numeric_formulas[[model_name]],
      data = data_numeric[-i, ]
    )

    pred_log <- predict(
      fit,
      newdata = data_numeric[i, , drop = FALSE]
    )

    # Back-transform using this training fold's residuals
    correction <- mean(exp(residuals(fit)))
    predictions[i] <- exp(pred_log) * correction
  }

  numeric_results[[model_name]] <- tibble(
    model = model_name,
    RMSE = sqrt(mean((data_numeric$m1 - predictions)^2)),
    MAE = mean(abs(data_numeric$m1 - predictions))
  )

  numeric_fits[[model_name]] <- lm(
    numeric_formulas[[model_name]],
    data = data_numeric
  )
}

numeric_comparison <- bind_rows(numeric_results) %>%
  arrange(RMSE)

numeric_comparison
```

```
# A tibble: 6 x 3
```

```
  model      RMSE     MAE
  <chr>     <dbl>   <dbl>
1 Re_Fr    0.0102 0.00521
2 pairwise 0.0106 0.00554
3 threeway 0.0112 0.00564
4 additive 0.0121 0.00666
5 St_Re    0.0122 0.00665
6 St_Fr    0.0123 0.00680
```

```r
library(glmnet)
```

```
Loading required package: Matrix


Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

    expand, pack, unpack

Loaded glmnet 5.1
```

```r
# Candidate terms for lasso
lasso_formula <- ~ log_St * log_Re * logistic_Fr +
  I(log_St^2) + I(log_Re^2) + I(logistic_Fr^2)

# Same predictor columns for every fold
x <- model.matrix(
  lasso_formula,
  data = data_numeric
)[, -1, drop = FALSE]

y <- data_numeric$log_mean

lasso_predictions <- numeric(nrow(data_numeric))

# Choose one penalty rule before running
penalty_rule <- "lambda.1se"  # Or "lambda.min"

set.seed(2026)
```

```r
# Outer LOOCV: hold out one observation
for (i in seq_len(nrow(data_numeric))) {

  x_train <- x[-i, , drop = FALSE]
  y_train <- y[-i]
  x_test  <- x[i, , drop = FALSE]

  # Inner 10-fold CV: choose the lasso penalty
  fold_id <- sample(
    rep(1:10, length.out = length(y_train))
  )

  fit <- cv.glmnet(
    x = x_train,
    y = y_train,
    alpha = 1,
    foldid = fold_id,
    standardize = TRUE
  )

  # Predict the held-out observation on the log scale
  pred_log <- as.numeric(
    predict(fit, newx = x_test, s = penalty_rule)
  )

  # Training residuals for the back-transformation correction
  fitted_log <- as.numeric(
    predict(fit, newx = x_train, s = penalty_rule)
  )

  correction <- mean(exp(y_train - fitted_log))

  lasso_predictions[i] <- exp(pred_log) * correction
}

# Original-scale errors, matching your ordinary regressions
lasso_results <- tibble(
  model = paste0("lasso_", penalty_rule),
  RMSE = sqrt(mean((data_numeric$m1 - lasso_predictions)^2)),
  MAE = mean(abs(data_numeric$m1 - lasso_predictions))
)
```

```r
# Compare with your existing models
all_comparison <- bind_rows(
  numeric_comparison,
  lasso_results
) %>%
  arrange(RMSE)

all_comparison
```

```
# A tibble: 7 x 3
  model               RMSE     MAE
  <chr>              <dbl>   <dbl>
1 Re_Fr             0.0102 0.00521
2 pairwise          0.0106 0.00554
3 threeway          0.0112 0.00564
4 additive          0.0121 0.00666
5 St_Re             0.0122 0.00665
6 St_Fr             0.0123 0.00680
7 lasso_lambda.1se 0.0142 0.00759
```

```r
library(glmnet)

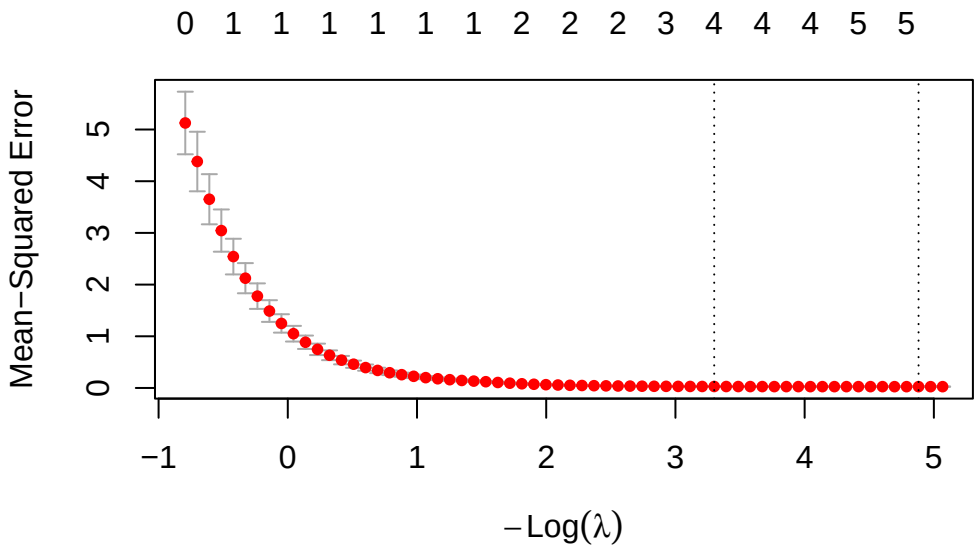
# Candidate predictors: main effects, two-way interactions,and quadratic terms to allow curva
x <- model.matrix(
  ~ log_St * log_Re * logistic_Fr +
    I(log_St^2) + I(log_Re^2) + I(logistic_Fr^2),
  data = data_numeric
)[, -1]

y <- data_numeric$log_mean

# Choose the penalty using 10-fold cross-validation
set.seed(2026)

lasso_cv <- cv.glmnet(
  x = x,
  y = y,
  alpha = 1,              # Lasso
  nfolds = 10,
  standardize = TRUE
)
```

```r
plot(lasso_cv)
```




```r
# Coefficients at the penalty with lowest CV error
coef(lasso_cv, s = "lambda.min")
```

```
11 x 1 sparse Matrix of class "dgCMatrix"
                             lambda.min
(Intercept)                15.327219735
log_St                      0.114464601
log_Re                     -3.893583087
logistic_Fr                 .
I(log_St^2)                 .
I(log_Re^2)                 .
I(logistic_Fr^2)            .
log_St:log_Re               0.003450178
log_St:logistic_Fr          0.080682071
log_Re:logistic_Fr          0.013979661
log_St:log_Re:logistic_Fr   .
```

```r
# More strongly penalized, usually simpler alternative
coef(lasso_cv, s = "lambda.1se")
```

```
11 x 1 sparse Matrix of class "dgCMatrix"
                             lambda.1se
(Intercept)                1.495536e+01
log_St                     1.523218e-05
log_Re                    -3.815783e+00
logistic_Fr                .
I(log_St^2)                .
I(log_Re^2)                .
I(logistic_Fr^2)           .
log_St:log_Re              2.702515e-02
log_St:logistic_Fr         3.095760e-02
log_Re:logistic_Fr         .
log_St:log_Re:logistic_Fr  .
```

```r
candidate_mean <- numeric_fits[["Re_Fr"]]

summary(candidate_mean)
```

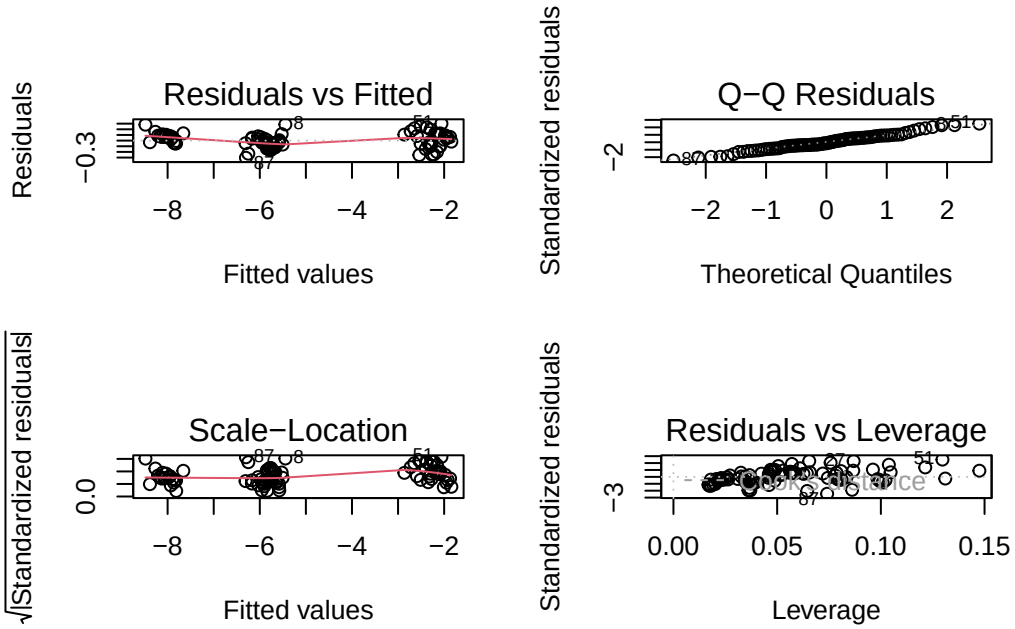
```
Call:
lm(formula = numeric_formulas[[model_name]], data = data_numeric)

Residuals:
     Min       1Q   Median       3Q      Max
-0.30375 -0.07027 -0.01734  0.08035  0.30050

Coefficients:
                   Estimate Std. Error t value Pr(>|t|)
(Intercept)        17.66991    0.41661  42.414  < 2e-16 ***
log_St              0.19627    0.01277  15.367  < 2e-16 ***
log_Re             -4.33391    0.07907 -54.811  < 2e-16 ***
logistic_Fr        -3.23271    0.57088  -5.663 2.03e-07 ***
log_Re:logistic_Fr  0.62006    0.10757   5.764 1.32e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1291 on 84 degrees of freedom
Multiple R-squared:  0.9968,    Adjusted R-squared:  0.9967
F-statistic:  6596 on 4 and 84 DF,  p-value: < 2.2e-16
```

```
par(mfrow = c(2, 2))
plot(candidate_mean)
```




```
par(mfrow = c(1, 1))
```

```
# Rank observations by influence
influence_table <- tibble(
  row = rownames(model.frame(candidate_mean)),
  cooks_distance = cooks.distance(candidate_mean),
  leverage = hatvalues(candidate_mean),
  studentized_residual = rstudent(candidate_mean)
) %>%
  arrange(desc(cooks_distance))

head(influence_table, 10)
```

```
# A tibble: 10 x 4
   row   cooks_distance leverage studentized_residual
   <chr>          <dbl>    <dbl>                <dbl>
 1 51             0.186    0.130                 2.58
 2 27             0.0991   0.0869                2.34
```

```
 3 87            0.0957   0.0741              -2.52
 4 8             0.0737   0.0655               2.35
 5 82            0.0717   0.104                1.78
 6 79            0.0685   0.0758               2.08
 7 86            0.0662   0.0861              -1.90
 8 21            0.0574   0.0645              -2.08
 9 17            0.0516   0.121                1.37
10 20            0.0424   0.0571               1.90
```

```
# Screening threshold-not an automatic deletion rule
4 / nobs(candidate_mean)
```

```
[1] 0.04494382
```

```
# Observations above the screening threshold
flagged_rows <- influence_table %>%
  filter(cooks_distance > 4 / nobs(candidate_mean)) %>%
  pull(row)

# Inspect their original values
data_numeric[
  match(flagged_rows, rownames(data_numeric)),
  c("St", "Re", "Fr", "m1")
]
```

```
# A tibble: 9 x 4
     St    Re      Fr       m1
  <dbl> <dbl>   <dbl>    <dbl>
1  3       90 Inf     0.172
2  0.1    398   0.052 0.000275
3  0.05   224   0.3   0.00135
4  3      224 Inf     0.00575
5  0.05    90   0.052 0.0879
6  0.1     90   0.052 0.105
7  0.05   224 Inf     0.00153
8  0.1     90   0.3   0.0613
9  0.1     90 Inf     0.0772
```

```
# Refit the same model, omitting one flagged observation at a time
sensitivity_fits <- setNames(
  lapply(flagged_rows, function(id) {
```

```
    reduced_data <- data_numeric[
      rownames(data_numeric) != id, ,
      drop = FALSE
    ]

    lm(formula(candidate_mean), data = reduced_data)
  }),
  paste0("without_", flagged_rows)
)

# Coefficient changes measured in full-model standard errors
full_coef <- coef(candidate_mean)
full_se <- sqrt(diag(vcov(candidate_mean)))

changes_in_se <- sapply(sensitivity_fits, function(fit) {
  (coef(fit)[names(full_coef)] - full_coef) / full_se
})

round(changes_in_se, 2)
```

```
                  without_51 without_27 without_87 without_8 without_82
(Intercept)             0.43       0.36      -0.08     -0.02      -0.28
log_St                 -0.48       0.41      -0.61     -0.42       0.44
log_Re                 -0.40      -0.39       0.09      0.04       0.27
logistic_Fr            -0.60      -0.26       0.04      0.01       0.22
log_Re:logistic_Fr      0.55       0.29      -0.06     -0.05      -0.21
                  without_79 without_86 without_21 without_17
(Intercept)            -0.32       0.00       0.24       0.24
log_St                  0.36      -0.45      -0.36       0.23
log_Re                  0.30      -0.03      -0.22      -0.22
logistic_Fr             0.25      -0.02      -0.16      -0.32
log_Re:logistic_Fr     -0.24       0.05       0.15       0.30
```

Nine observations exceeded the Cook's distance screening threshold. Removing each individually changed all coefficients by less than 0.62 full-model standard errors, suggesting that the coefficient estimates were reasonably robust to individual influential observations.

```
summary(candidate_mean)
```

```
Call:
```

```
lm(formula = numeric_formulas[[model_name]], data = data_numeric)

Residuals:
     Min       1Q   Median       3Q      Max
-0.30375 -0.07027 -0.01734  0.08035  0.30050

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)       17.66991    0.41661  42.414  < 2e-16 ***
log_St             0.19627    0.01277  15.367  < 2e-16 ***
log_Re            -4.33391    0.07907 -54.811  < 2e-16 ***
logistic_Fr       -3.23271    0.57088  -5.663 2.03e-07 ***
log_Re:logistic_Fr  0.62006    0.10757   5.764 1.32e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1291 on 84 degrees of freedom
Multiple R-squared:  0.9968,    Adjusted R-squared:  0.9967
F-statistic:  6596 on 4 and 84 DF,  p-value: < 2.2e-16
```

```
confint(candidate_mean)
```

```
                       2.5 %     97.5 %
(Intercept)       16.8414308 18.4983805
log_St             0.1708718  0.2216689
log_Re            -4.4911520 -4.1766710
logistic_Fr       -4.3679658 -2.0974526
log_Re:logistic_Fr  0.4061521  0.8339771
```

Mean cluster volume increased with St, with doubling St associated with approximately 14.6% higher fitted volume, holding Re and Fr fixed. Higher Re was associated with lower volume across the observed gravity settings. The positive Re–Fr interaction indicated that this relationship became less negative as Fr increased.